

SaaS Product Engagement & Expansion Intelligence

Day 1 — Project Initiation

Work Done:

Reviewed the complete project blueprint and confirmed the SaaS Customer Journey and Operational Insights dataset. Assessed the source credibility, licensing, and freshness. Created the GitHub repository with separate folders for `data`, `sql`, `notebooks`, `dashboard`, and `docs`.

Learnings:

Learned how to understand project requirements before implementation and how to establish a professional repository structure. Also understood the importance of checking dataset reliability and planning the project workflow from the beginning.

Day 2 — Problem Definition & Data Dictionary

Work Done:

Restated the business problem and success metrics. Identified `product_operations.csv` and `customer_profile.csv` as the main datasets and mapped the approved columns to their business meanings. Created the first version of the data dictionary.

Learnings:

Learned how to translate technical dataset fields into business terminology. Understood how a data dictionary improves consistency and makes the dataset easier to understand during analysis.

Day 3 — Data Profiling

Work Done:

Checked the row and column counts of both datasets, inspected data types, recorded missing-value counts, and identified potential duplicate records using `record_id` and `customer_id`.

Learnings:

Learned how to systematically profile a dataset before cleaning. Understood how missing values, duplicate records, and incorrect data types can affect the reliability of later analysis.

Day 4 — Data Quality Assessment

Work Done:

Identified numeric-as-text issues, malformed subscription dates, invalid or out-of-range values in financial fields, and inconsistent categories in `plan_category`, `payment_status`, and `subscription_status`. Documented the findings in the profiling report.

Learnings:

Learned how to identify different types of data-quality problems and understand their effect on analysis. Improved my understanding of validating numerical, date, and categorical fields.

Day 5 — Cleaning Plan**Work Done:**

Prepared a formal cleaning plan covering missing values, duplicate records, data-type corrections, and category harmonization. Defined whether affected fields should be kept, dropped, or imputed and finalized Cleaning Plan v1.

Learnings:

Learned that data cleaning should follow clear and documented rules. Understood how a predefined cleaning plan improves consistency, reproducibility, and data-quality management.

Day 6 — Excel Data Cleaning**Work Done:**

Opened both datasets in Excel and used filters and conditional formatting to inspect the data. Applied the cleaning plan by correcting data types, trimming whitespace, and harmonizing plan, payment, and subscription categories. Reconciled cleaned row counts with the original files.

Learnings:

Improved practical Excel cleaning skills and learned how to use filters and conditional formatting for data-quality checks. Also learned the importance of validating row counts after cleaning.

Day 7 — Excel Analysis**Work Done:**

Created pivot tables based on `plan_category`, `payment_status`, and `subscription_status`. Analyzed `discount_percent`, `tax_amount`, and `final_amount_paid`, created initial summaries, and identified the first set of business insights.

Learnings:

Learned how pivot tables can quickly summarize large amounts of business data. Improved my ability to compare numerical metrics across different business categories and identify initial patterns.

Day 8 — SQL Database Setup**Work Done:**

Loaded the Product Operations and Customer Profile datasets into SQL and defined keys using `record_id` and `customer_id`. Reproduced the cleaning process in SQL, including null handling, deduplication, and data-type casting. Created the database schema.

Learnings:

Learned how to move prepared data into a relational database and reproduce data-cleaning logic using SQL. Improved my understanding of keys, table structures, and maintaining data integrity.

Day 9 — SQL Joins & KPI Queries**Work Done:**

Created SQL joins between the two datasets and performed aggregations on discount, tax, and final payment values. Grouped the results by plan category, payment status, and subscription status and developed KPI queries based on the project requirements.

Learnings:

Improved my understanding of SQL joins, grouping, and aggregation. Learned how to convert business requirements into structured SQL queries and how combining datasets provides a broader analytical view.

Day 10 — SQL Views & Window Functions**Work Done:**

Created SQL views for the main KPIs and used window functions with subscription dates for running totals, ranking, `LAG`, and `LEAD` analysis. Organized the SQL files and added documentation.

Learnings:

Learned how SQL views simplify repeated analytical queries and improved my understanding of window functions. Learned how ranking and previous/next-period comparisons can support time-based business analysis.

Day 11 — Python Cleaning & Validation**Work Done:**

Reproduced the cleaning process using Pandas in Jupyter Notebook. Validated row counts and null counts against the cleaning plan using assertions and committed the cleaning notebook.

Learnings:

Learned how to implement the same cleaning methodology programmatically using Pandas. Improved my understanding of DataFrame validation and learned how assertions can help detect unexpected changes in the dataset.

Day 12 — Exploratory Data Analysis**Work Done:**

Performed EDA on `discount_percent`, `tax_amount`, and `final_amount_paid`. Generated summary statistics and analyzed the data by plan category, payment status, and subscription status. Derived date components from subscription start and end dates.

Learnings:

Learned how EDA helps identify patterns before visualization. Improved my skills in Pandas aggregation, segmentation, date manipulation, and numerical distribution analysis.

Day 13 — Outlier & Visualization Analysis**Work Done:**

Performed outlier analysis using IQR and Z-score methods on discount, tax, and final payment values. Created distribution, trend, comparison, and composition charts and documented five dataset-grounded insights.

Learnings:

Learned how statistical methods can identify unusual observations and why outliers should be investigated before interpreting business metrics. Improved my understanding of choosing charts according to the analytical question.

Day 14 — Descriptive Statistics**Work Done:**

Calculated descriptive statistics for discount percentage, tax amount, and final payment. Analyzed distributions using histograms, skewness, and kurtosis and documented sampling-related caveats.

Learnings:

Learned how descriptive statistics explain the central tendency and spread of data. Improved my understanding of skewness and kurtosis and how distribution characteristics influence data interpretation.

Day 15 — Statistical Analysis**Work Done:**

Created a correlation matrix for discount, tax, and final payment values. Performed variance analysis across plan category, payment status, and subscription status. Statistically validated the main hypothesis and documented the limitations.

Learnings:

Learned how correlation and variance analysis can be used to investigate relationships and differences across business categories. Also learned that statistical relationships should be interpreted carefully and do not automatically establish causation.

Day 16 — Power BI Model & KPIs**Work Done:**

Imported the cleaned data into Power BI and defined relationships between the Product Operations and Customer Profile datasets. Created DAX measures and KPI cards for the main business metrics.

Learnings:

Learned how to build a Power BI data model and create relationships between datasets. Improved my understanding of DAX and how reusable measures can be used to calculate business KPIs.

Day 17 — Dashboard Development**Work Done:**

Built the main Power BI pages: **Overview, Trends, and Segments**. Added slicers for plan category, payment status, and subscription status and implemented drill-down functionality from summary to detail.

Learnings:

Learned how to structure a multi-page interactive dashboard. Improved my understanding of slicers, drill-downs, visual organization, and designing dashboards around specific business questions.

Day 18 — Dashboard Design & Storytelling**Work Done:**

Applied business-focused titles, annotations, and callouts to the dashboard. Improved theme consistency, spacing, tooltips, and accessibility. Exported the final .pbix file and dashboard screenshots to the repository.

Learnings:

Learned that dashboard development is not only about calculations and charts. Clear titles, spacing, annotations, tooltips, accessibility, and visual storytelling are equally important for communicating insights effectively.

Day 19 — Business Insights & Recommendations**Work Done:**

Finalized business insights and recommendations linked to the dashboard KPIs. Documented dataset limitations and created the cleaning log containing the cleaning actions and validation details.

Learnings:

Learned how to convert analytical findings into business-oriented insights and recommendations. Understood that recommendations should be directly supported by data and that limitations should be clearly communicated.

Day 20 — Documentation & Portfolio

Work Done:

Finalized the Data Dictionary and README. Added purpose comments to SQL and Python files and prepared dashboard documentation covering measure definitions and page purposes. Organized the required portfolio artifacts.

Learnings:

Learned how to document a complete analytics project professionally. Improved my ability to explain the purpose of technical files, dashboard measures, page structure, and overall methodology.

Day 21 — Final Presentation & Submission

Work Done:

Rehearsed the final presentation covering both the business and technical narrative. Prepared the technical defence for data cleaning, SQL, statistics, Python, and Power BI. Reviewed the final project folder, GitHub repository, dashboard, documentation, and portfolio links before submission.

Learnings:

Learned how to present a technical project as a complete business story. Improved my ability to explain technical decisions, defend analytical methods, answer project-related questions, and connect the complete workflow from **raw data to business insights**.